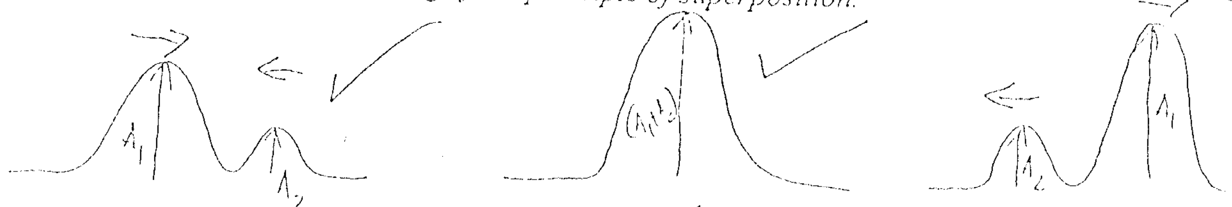
VRAAG / *QUESTION* 5

(a) Verduidelik skematies wat met die superponeringsbeginsel bedoel word.
Show schematically the meaning of the principle of superposition. (2



When 2 sinusoidal waves meet, the will produce a wave which is the algebraic sum of the 2 separate waves.

(2)

(b) Twee golwe $y_1(x,t) = Y \text{ Sin } (kx - \omega t)$ en $y_2(x,t) = Y \text{ Sin } (kx - \omega t + \phi)$ word gesuperponeer.
Two waves $y_1(x,t) = Y \text{ Sin } (kx - \omega t)$ en $y_2(x,t) = Y \text{ Sin } (kx - \omega t + \phi)$ *are superimposed.*

Gegee / *Given:* $\text{Sin } A + \text{Sin } B = 2 \text{ Sin } \frac{A+B}{2} \text{ Cos } \frac{A-B}{2}$

(i) Toon aan dat die amplitude van die nuwe golf is:
Show that the amplitude of the new wave is:

$$Y' = 2Y \text{ Cos } \frac{1}{2}\phi$$

NB $(\cos(-\frac{\phi}{2}) = c$

$y_1 + y_2 = y\sin(kx - \omega t) + y\sin(kx - \omega t + \phi)$

$= 2y\left(\sin\frac{(kx-\omega t + kx - \omega t + \phi)}{2}\right)\left(\cos\frac{kx-\omega t - kx + \omega t - \phi}{2}\right)$

$= \underbrace{2y\left(\cos\frac{-\phi}{2}\right)}\left(\sin\left(kx - \omega t + \frac{\phi}{2}\right)\right)$ (4

∴ the new amplitude is $2Y\cos\left(\frac{1}{2}\phi\right)$

(4)

(ii) Wanneer konstruktiewe interferensie plaasvind is die fase:
For constructive interference the phase is:

$\phi = n(2\pi)$ $n = 0, 1, 2, 3, \ldots$

en die nuwe amplitude word bepaal as: / *and the new amplitude can be determined as:*

$Y' = 2Y$ $(\cos\pi = -1$ etc...$)$ (4

Vir destruktiewe interferensie is die fase: / *For destructive interference the phase is:*

$\phi = (n + \frac{1}{2})2\pi$ $n = 0, 1, 2, 3, \ldots$

en die amplitude is: / *and the amplitude will be:*

$Y' = 0$

(4)

Vraag / *Question* 5(c)(Vervolg / *Continue*)/....